

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHM 319 — Past Questions & Materials
by Renance

Course: CHM 319
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

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Paper 1 of 1 — CHM 319 - 2024/2025 - First Semester - APPLIED SPECTROSCOPY

Level 200 • First Semester • PastQuestion

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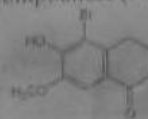
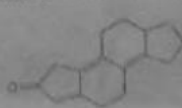
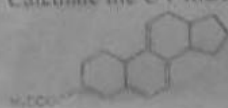
FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
DEPARTMENT OF CHEMISTRY
FIRST SEMESTER EXAMINATIONS 2024/ 2025 SESSION

CHM 319 APPLIED SPECTROSCOPY TIME: 2 1/2 HRS

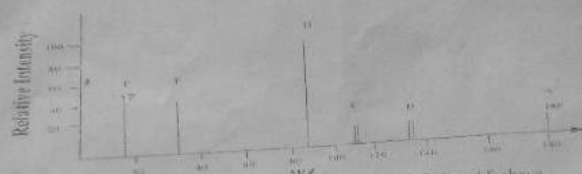
Answer question 1 and any other three questions

1. The IR, UV, MS ^1H and ^{13}C NMR of an organic compound are provided in the sheet labeled Y. Show all possible deductions from the information in the spectral sheet provided and propose a molecular structure for the compound. (25 mks)

- 2(a). Define the following terms as applicable to UV spectroscopy (2 mks). ✓
(i). Chromophore (ii). Auxochrome
- 2(b). List three (4) characteristics of solvents used in Ultraviolet spectroscopy (4 mks).
- 2(c). Calculate the UV maximum for the following compounds. (9 marks)



- 3a) Describe the following concepts in terms of vibrational frequency. (i) Bond length (ii) Mass of an atom. (2mks)
- 3b) Arrange the following functional groups in the order of highest frequencies of vibration, -COOR, -OH, -CO- and -CHO. (1mk)
- 3c) The vibration of atoms in infrared spectroscopy is caused by -? (1mk) ✓
- 3d) List two types of infrared spectroscopy? (2mks) ✓
- 3e) Give a brief explanation of the three regions in the correlation chart of an IR spectra. (3mks)
- 3f) List two advantages of FTIR. (2mks)
- 3g) Draw a schematic diagram of a dispersive infrared spectrometer and label. (3mks)
- 3h) What is the basic principle in infrared spectroscopy? (1mk)
- 4a) Differentiate between GC-MS and LC-MS. (4mks) ✓
- 4b) With schematic diagram, briefly explain the instrumentation of MS. (3mks) ✓
- 4c) State three applications of MS. (3mks) ✓

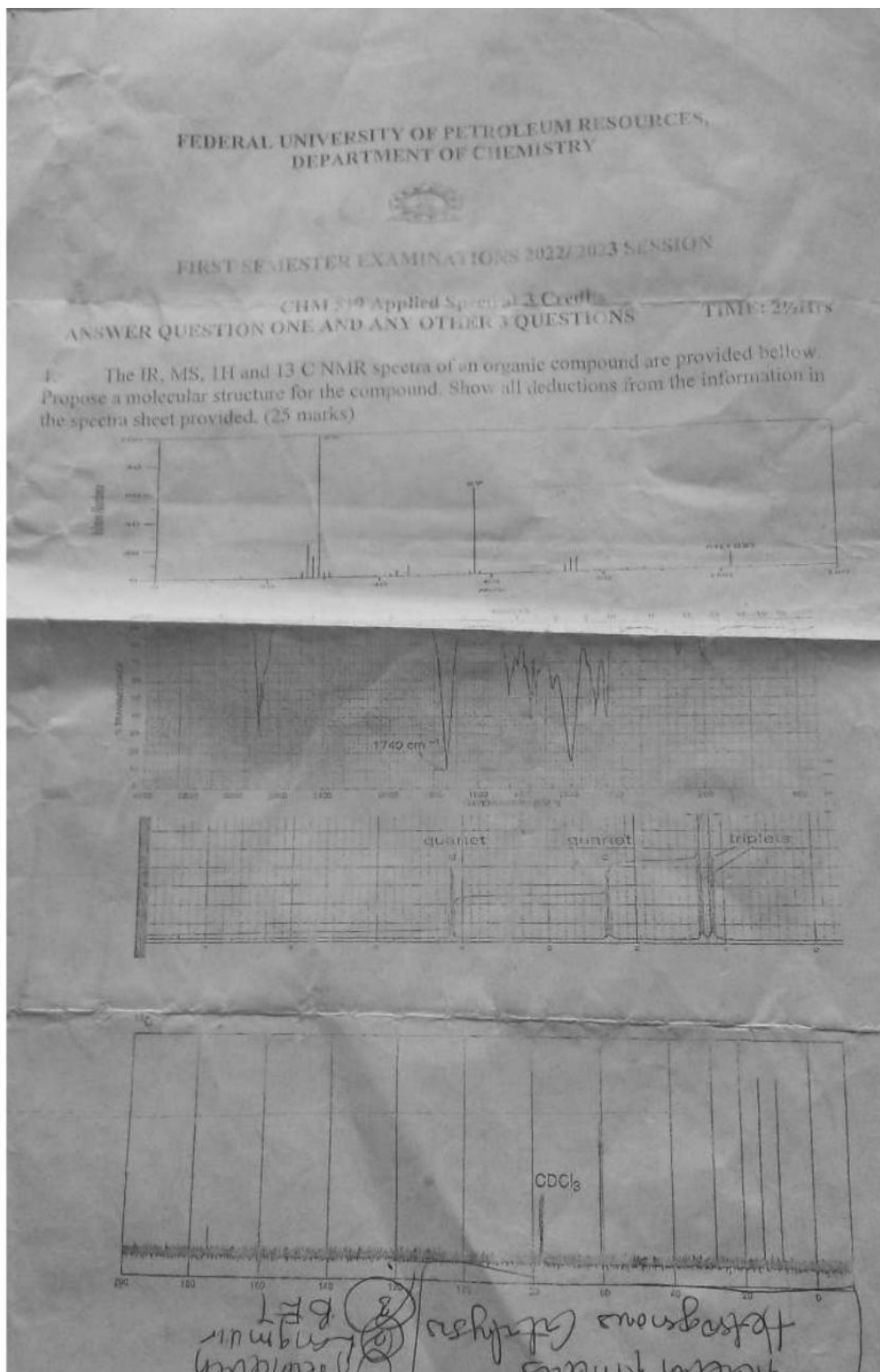


- 4b) Consider the MS Spectrum with peaks lettered A, B, C, D, E and F above.
Which peaks in the mass spectrum corresponds to the molecular ion (M^+) and what is its mass-to-charge (m/z) ratio? (2mks)
- 4c) Which peak in the spectrum is the base peak? (1mk)
- 4d) Give the structure and mass of any two characteristic fragments in the mass spectrum of the ethyl benzene (2mks)
- 5a) List any four (4) atoms that will be active in the NMR region (2mks)
- 5b) For each of the following compounds, give number of proton signals expected and number of carbon signals expected (2mks)
- (i) $CH_3CH_2CH_2Br$ (ii) $CH_3CH_2OCH_2CH_2CH_2COOCH_2CH_3$ (8mks)
- 5c) Using your data sheet give the approximate chemical shift position of the carbon atoms in methyl cyclohexane (2mks)
- 5d) Explain any one factor that affects J-value (2mks)

END OF PAPER

C	H	O
5.82	8.5	0
5.82	8.5	0
12	1	16
0.48	8.5	

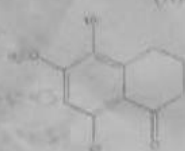




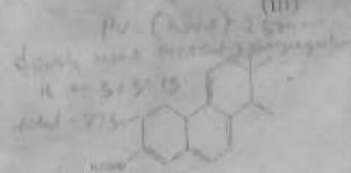
- 2(a). Define the following terms as applicable to UV spectroscopy.
 (i). Chromophore (ii). Auxochrome (iii). Hypsochromic shift (iv). Hyperchromic effect (6 marks).
 (b). Sketch and label the Infra-red spectrometer. (5 marks)
 (c). List three (3) characteristics of solvents used in Ultraviolet spectroscopy (3 marks)
 (d). Calculate the index of Hydrogen deficiency for the compounds below.
 $C_8H_8Cl_2$ (3 marks)

3(a). Calculate the UV wavelength for the following compounds. (2 marks)

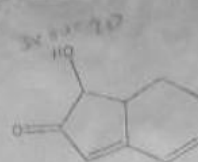
(i).



(ii).



(iii).



- (e). Discuss extensively, the three (3) methods of handling solid samples in Infra-red spectroscopy (5 marks).
 (d). List two uses three (3) uses of IR spectroscopy (3 marks).

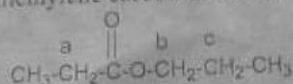
4. Outline three sequences of what happens in a mass spectrometer (4mks).

5c. Sketch and label a simple mass spectrometer (4mks)

4d. Five peaks in the mass spectrum shows that there are five isotopes of Zirconium with the relative isotopic masses of 90, 91, 92, 94 and 96 on a ^{12}C scale. What is the relative atomic mass of Zirconium? (3mks)

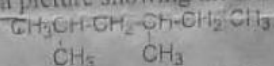
TOTAL = 17mks

5a) Match the ^{13}C -NMR chemical shift values (ppm) on the right with the lettered methylene carbon atoms on the left. (3mks)



68.9 22.7 26.4

- 5b) i. Draw a picture showing the 1H -NMR splitting patterns (singlet, doublet, etc.) for each methylene group in the structure above. (6mks)
 ii. Draw a picture showing the DEPT 135 Spectrum for the structure below. (2mks)



- 5c). Draw an expanded structural formula for each compound below. How many different resonance peaks would you expect to find in the 1H -NMR spectrum of each of the compounds (ignore spin-spin splitting). (6mks)
 i. ethyl methyl ether ii. 1-nitropropane iii. Bromoethane

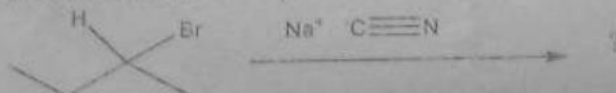
Chromophore is a molecule that is responsible for color. It absorbs light in the visible region. Auxochrome - This is a functional group that when attached to a chromophore, it extends the chromophore's ability to absorb light.

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
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FIRST SEMESTER EXAMINATIONS 2013/ 2014 SESSION

CHM 411 Reaction Mechanism

ANSWER 4 QUESTIONS ONLY TIME: 2 1/2 Hrs

- 1a. (i) Give the structure of any compound with three asymmetric carbon atoms (1mk)
(ii) How many stereoisomers will be produced from (1a) above? (1mk)
(iii) Draw the structures of a pair of possible diastereomers from (1a) above (2mks)
(b) Give the structure of the product of the reaction below under S_N2 conditions (2mks)

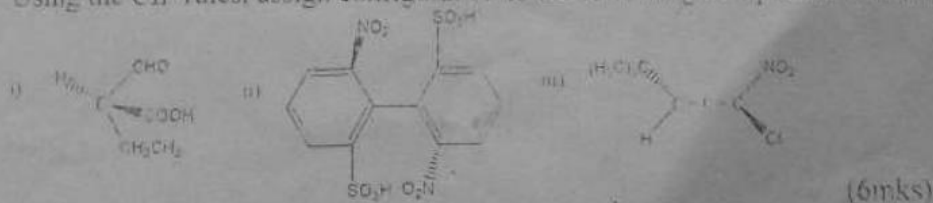


- 1c. What is a racemic mixture? Give the explanation to an enantiomeric excess of 25% of the positive enantiomer in a solution. (4mks)
1d. With the aid of three suitable examples, differentiate between stereospecific and stereoselective reactions. (3mks)
1e. Using chemical equations only outline two examples of a regioselective reaction. (4mks)

- 2a. What is chirality? Draw the structures of two chiral and two achiral compounds. (6mks) ✓
2b. Deduce the structures of any geometrical conformer of your choice with the following configuration (i). Trans, Cis (ii) Cis, Trans (iii) Cis, Cis (iv) Trans, Trans (4mks) ✓
2c. Define the following terms and illustrate with an example.
(i) Enantiomers (2mk) (ii) Meso compound (2mk) ✓
2d. Define Diastereomers? Give two physical properties that can be used to differentiate diastereomers (3mks) ✓

- 3a. Explain two methods of resolving racemic mixtures. (4mks)
3b. Mention the two types of stereoisomerism and explain them? (4mks)
3c. Give two importance of stereoisomeric reactions (2mks)
3d. What instrument is used to measure the rotatory power of a substance? Describe the instrumentation setup (4mks)
3e. Outline three conditions to determine specific configuration of R and S. (3mk) - Chirality

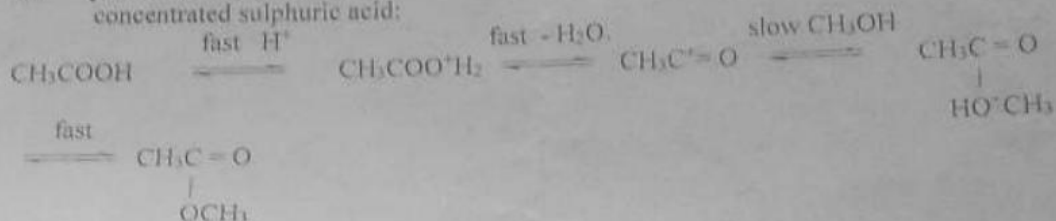
- 4(a) Explain any two factors that can lead to optical activity in a molecule. (4 mks)
4(b) Will 4,4' biphenyl be optically active? Give reason for your answer (3mks) ✓
4c Using the CIP rules, assign configuration to the following compounds showing all steps



- 4d Draw the structure of (i) (1R,3S)-1,3-dichlorocyclopentane
(ii) (2R,3S)-3-chlorobutan-2-ol (4mks)

- 5a What is the difference between a conformation and a configuration? (2mks)
5b Explain any two factors that affect the energy content of conformers (4mks)
5c List three characteristics of groups for neighbouring group participation in S_N1 (3mks)
5d Explain how the use of isotopes in determination of reaction mechanism (6mks)
5e What is element effect in determination of reaction mechanism? Give example (2½mks)

- 6a Given the mechanism for the reaction of ethanoic acid with methanol in the presence of concentrated sulphuric acid:



- (i) What is the order of the reaction? *first*
(ii) What is the molecularity of the reaction? *uni*
(iii) If the methanol is changed to isopropanol, what do you expect will happen to the rate of reaction? Give your reasoning.
(iv) If the first step is the slowest step, what will be the order of reaction? Why? (2mks)

- 6b In the reaction $\text{C}_6\text{H}_6 + \text{CH}_3\text{AlCl}_2 \longrightarrow \text{C}_6\text{H}_5\text{CH}_3 + \text{AlCl}_3 + \text{HCl}$
Explain the effect of the presence of $-\text{NO}_2$ and $-\text{OCH}_3$ respectively on the rate of reaction and types of product isomers formed. (7½mks)

END OF PAPER